SAMAR ALI

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RESEARCH FOCUS

Speech enhancement, source separation, target-speaker extraction, beamforming, robust ASR, and efficient multimodal systems deployed on edge hardware.

EDUCATION

B.Tech, Electronics & Communication Engineering (VLSI Design & Technology) · Delhi Technological University, New Delhi — *2025 – Present · 8.0/10 CGPA*

EXPERIENCE

Software Engineer · UAS-DTU (Unmanned Aerial Systems) — *Sep 2025 – Present*

- Develop the speech and audio subsystem for autonomous triage robots, deploying vision, audio, and language models on UAV/UGV edge compute.
- Built the speech pipeline behind a prize-winning DARPA Triage Challenge system, combining multichannel processing, neural enhancement, ASR, TTS, and a fine-tuned language model.
- Developed a UAV-swarm control and monitoring GUI and tuned Pixhawk/ArduPilot flight controllers for field operations.

Co-Head · AIMS-DTU (AI & Machine Learning Society) — *Oct 2025 – Present*

- Lead applied AI and computer-vision work while organizing workshops, hackathons, and technical sessions.
- Mentor students through practical machine-learning projects and structured hands-on sessions.

SELECTED PROJECTS

Direction-Conditioned Target Speaker Extraction · PyTorch, complex U-Net, MVDR

- Built a direction-conditioned neural beamformer that extracts a target speaker from an 8-microphone mixture; improved SI-SDR from 0.6 dB to 11.2 dB for voices separated by 3° and achieved 10.94 dB validation SI-SDR.

Custom 3D Microphone Array · Raspberry Pi Pico 2, I2S MEMS microphones

- Built an 8-channel array on a 3D-printed frame, writing sample-synchronized firmware and a host-side driver for plug-and-play USB capture.

Isaac Newton Digital Twin · RAG, Chroma, Whisper, TTS

- Developed a voice-enabled historical digital twin with semantic and keyword retrieval, long/short-term memory, two personas, and four interactive simulations.

OmniControl Smart Glove · ESP32, MediaPipe, BLE, Wi-Fi UDP

- Built a multimodal wearable controller mapping 10 hand gestures to system actions, with IMU cursor control, touch input, Wi-Fi UDP, and BLE fallback.

TECHNICAL SKILLS

Languages: Python, JavaScript, C/C++, HTML/CSS

Speech/Audio: MVDR, GSC, DOA, dereverberation, BSS, GSS, TSE, ASR, TTS, VAD, SpeechBrain, torchaudio, Whisper

ML/AI: PyTorch, TensorFlow, Hugging Face, scikit-learn, OpenCV, MediaPipe, RAG, edge deployment

Hardware: Pixhawk/ArduPilot, NVIDIA Jetson, Raspberry Pi, Pico 2, ESP32, I2S MEMS microphones

Tools: NumPy, SciPy, Pandas, Git/GitHub, Docker, Podman, Linux, React

CERTIFICATIONS

- Machine Learning Specialization · Coursera / Stanford
- Deep Learning Specialization · DeepLearning.AI
- Convolutional Neural Networks · DeepLearning.AI
- Sequence Models · DeepLearning.AI

View certificates

ACHIEVEMENT

- Prize-winning system contributor, DARPA Triage Challenge, an international competition focused on medical triage and autonomous robotics.